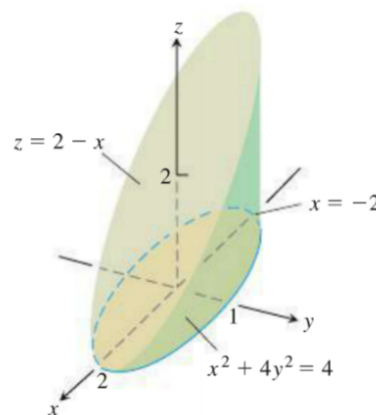


Math 2551 Worksheet: Mass and Moments

1. Find the center of mass of a thin plate bounded by the line $y = 1$ and the parabola $y = x^2$ if the density is $\delta(x, y) = y + 1$. *Hint: Consider symmetry.*

2. A solid of constant density $\delta(x, y, z) = 2$ is bounded below by the plane $z = 0$, on the sides by the elliptical cylinder $x^2 + 4y^2 = 4$, and above by the plane $z = 2 - x$. Set up all of the necessary triple integrals to compute its center of mass. You do not need to compute the center of mass.



Useful Formulas

- Mass: $M = \iiint_D \delta \, dV$ or $M = \iint_R \delta \, dA$
- First moments (2D plate): $M_y = \iint_R x\delta \, dA$, $M_x = \iint_R y\delta \, dA$
- Center of mass (2D plate): $(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M} \right)$
- First moments (3D solid): $M_{yz} = \iiint_D x\delta \, dV$, $M_{xz} = \iiint_D y\delta \, dV$, $M_{xy} = \iiint_D z\delta \, dV$
- Center of mass (3D solid): $(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_{yz}}{M}, \frac{M_{xz}}{M}, \frac{M_{xy}}{M} \right)$